

Attention-Based Deep Learning Framework for Skin Lesion Classification with Explainable AI

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This draft records the project's research basis, completed implementation, and validation findings for discussion with the project guide. The central method is an ImageNet pretrained EfficientNet B0 with a Convolutional Block Attention Module (CBAM) for seven class HAM10000 classification, with Grad CAM for visual explanation. The original lesion separated test partition is reserved; no final test accuracy is reported.

Abstract

Dermoscopic lesion classification is difficult because several conditions look similar and HAM10000 has severe class imbalance. We implemented an image only EfficientNet B0 plus CBAM pipeline and compared weighted cross entropy, random oversampling, focal loss, probability averaging, feature based classifiers, metadata fusion, and inference corrections. Model selection used macro F1 on a fixed lesion separated validation set. The original weighted run achieved 84.56% validation accuracy and 0.7569 macro F1. The highest lesion separated accuracy measured so far is 87.69%, from an exploratory PanDerm and EfficientNet ensemble; its highest macro F1 is a different configuration at 0.7837. A separate image level validation experiment reached 90.75% with an eight stream PanDerm and EfficientNet ensemble, but its training and validation sets share lesion identities. The strongest EfficientNet plus CBAM result on that exploratory split is 88.16%. These values describe distinct protocols and are not test results. The proposed final methodology remains within the EfficientNet, CBAM, and Grad CAM family; final configuration selection, explanation review, and held out testing remain pending.

1 Introduction

Skin lesions include benign conditions and malignancies. Automated analysis of dermoscopic photographs may help organize visual evidence, but a classifier cannot replace clinical assessment. HAM10000 provides seven diagnosis categories and repeated images of some lesions. Two research difficulties are central: the nevus class dominates the image count, and images of the same lesion can be visually similar. A high accuracy on a split that mixes those images can overstate performance on unseen lesions.

Our research question is whether an efficient CNN with learned channel and spatial attention can classify all seven categories while retaining useful minority class performance and offering understandable visual explanations. We also ask how balancing choices and evaluation protocol change the apparent result. The aim is a defensible research comparison, not only the largest percentage.

Problem Statement

The research problem is seven-way classification of dermoscopic images under severe class imbalance and visual similarity between diagnoses. A single dominant class can make overall accuracy look strong while melanoma or other less common categories are missed. More than one image may show the same lesion, creating a second challenge: an image-level random split can place related views on both sides of training and validation. We therefore need a model that reports performance across classes and an evaluation design that makes the source of each result clear.

The system accepts an image and returns scores for the seven HAM10000 categories. These scores are research predictions, not clinical diagnoses. Its effectiveness is judged on saved validation partitions; the original test partition is deliberately withheld until the final configuration is fixed. This boundary is important because testing candidate after candidate on the same supposed test set would make that set part of model selection.

Research Motivation

EfficientNet offers a practical pretrained visual backbone on the available 8 GB GPU. CBAM is a small attention block that can change which channel and spatial signals the classifier emphasizes, while Grad CAM can later show where a class score receives support. This combination matches the approved project direction and supports an interpretable analysis of errors. Class weighting and related training interventions address the very uneven HAM10000 counts without altering the original validation and test distributions.

Published HAM10000 percentages are often difficult to compare because investigators may use different class subsets, balance before or after a split, select different lesion populations, or report validation-selected results as if they were independent tests. A useful student research contribution is therefore a well documented sequence of decisions and outcomes. The objective is to explain why a technique helped, hurt, or produced a trade-off under a named protocol, instead of treating every larger percentage as a better model.

Objectives

- Build a reproducible seven class image classifier using an EfficientNet backbone and CBAM.
- Compare train only balancing and loss strategies with accuracy, per class precision and recall, macro F1, and confusion matrices.
- Keep lesion identities separate in the primary development protocol and reserve the test partition for final evaluation.
- Use Grad CAM to inspect what image regions influence predictions, while stating that a heatmap does not prove clinical reasoning.
- Document exploratory protocol and architecture probes separately from the proposed final method.

2 HAM10000 Dataset and Evaluation Design

HAM10000 contains 10,015 dermoscopic images from 7,470 lesion identities. Its class counts are nv 6,705, mel 1,113, bkl 1,099, bcc 514, akiec 327, vasc 142, and df 115. The seven codes denote melanocytic nevus, melanoma, benign keratosis like lesion, basal cell carcinoma, actinic keratosis or intraepithelial carcinoma, vascular lesion, and dermatofibroma. Image files supply the model input; metadata supplies labels and lesion IDs. Age, sex, and anatomical site are not used by the proposed image only model.

The primary fixed split uses stratified grouping by lesion ID, seed 42. It contains 7,009 training images, 1,503 validation images, and 1,503 locked test images, with zero lesion ID overlap among partitions. Training determines parameters, validation selects checkpoints and compares methods, and the test set is reserved for one final assessment after method lock. Selecting many variants on the same validation set can make the best validation score optimistic even without lesion leakage.

An additional exploratory image level split uses only the original train plus validation pool and keeps the original test images excluded. It has the same 7,009/1,503 development counts but 563 lesion identities cross the train/validation boundary; 596 validation images share a lesion with training. Its score is therefore labeled exploratory throughout. Different validation membership and lesion overlap prevent attributing its difference from the primary split to one cause alone.

Accuracy is the fraction of correct predictions. Macro precision, recall, and F1 average each class equally, making poor rare class performance harder to hide. We retain per class scores and confusion matrices. AUC ROC appeared in the original plan but is not reported here as a completed outcome. All results in this draft are validation results; no held out test result exists yet.

Class Imbalance and Dataset Challenges

The original class distribution ranges from 6,705 nevus images to 115 dermatofibroma images. A classifier trained on natural frequency alone can learn that predicting nevus is often rewarded, especially when measured only by accuracy. Macro F1 partly counters this by giving every class an equal vote in the average. It does not make every error equally serious in a medical sense, but it reveals whether a gain in accuracy comes with poor minority-class performance.

HAM10000 combines different sources, and images can differ in illumination, framing, artifacts, and lesion size. The baseline resize from 600 by 450 to 224 by 224 also changes aspect ratio and reduces detail. We tested one inference-only crop diagnostic, which harmed performance, but this does not prove that all geometry changes are unhelpful: a model trained and evaluated under a matched crop or resolution would be a distinct experiment. We avoid claiming that dataset noise or acquisition shift explains a precise fraction of the observed accuracy plateau because no such causal audit was completed.

Primary and Exploratory Splits

The primary grouped split separates all images with the same lesion ID before any balancing. It was generated once and then retained. This means the 1,503 validation images represent lesions not present in the 7,009-image training partition. The separate exploratory image-level split uses the same 8,512-image development pool and preserves the original 1,503 test images. Its 563 overlapping lesion IDs and 596 related validation images are documented rather than ignored. The two validation cohorts differ in membership as well as overlap, so a score difference alone cannot isolate the effect of leakage.

The validation set is used repeatedly for checkpoint selection and comparisons. Thus even its lesion-disjoint scores are development estimates, not a substitute for a final held-out test. Multiple exploratory grid searches can overfit model choice to validation. The report therefore records each protocol, cohort size, and paired accuracy and macro F1 result, and limits conclusions to what the saved evidence supports.

3 Literature Review

Study or source	Method and reported result	Influence and comparison
Tschandl et al. 2018 [1]	HAM10000 dataset: 10,015 multi source dermoscopic images.	Established seven class task and lesion ID grouping need. It is a dataset paper, not a model performance benchmark.
Pardede et al. 2026 [2]	EfficientNet B0 versus MobileNetV3 Large, oversampling and Grad CAM; 98.86% and 98.88% reported.	Base architecture and imbalance reference. Its Table IV totals 7,041 balanced test cases, so its test cohort is not an untouched 15% sample of the 10,015 originals; direct comparison is inappropriate.
Haque et al. 2026 [3]	EfficientNetV2 L with attention, 384 pixel input and staged training; 91.15% accuracy, 0.8545 macro F1.	Motivated resolution and augmentation checks. Much larger model, compute, and image count split; lesion overlap is not established.
PanDerm [4]	Dermatology pretrained vision transformer with released Base weights and probing recipe.	Motivated a separate domain feature probe. It is outside the agreed EfficientNet plus CBAM final architecture.
Applied Sciences ensemble 2024 [5]	Reported 96% on a balanced three category HAM10000 subset.	Shows class set and balancing can change the headline. It is not a seven class comparison.
Agra EfficientNet B2 project [6]	ImageNet EfficientNet B2 with augmentation and image level split; 82.24% test accuracy reported.	Its code informed a cautious augmentation probe and highlighted the need to inspect split code, not only reported numbers.

The base paper is particularly relevant because it combines an efficient pretrained backbone, an imbalance intervention, and a visual explanation. Its reported 98.86% is retained as the authors' result, not adopted as our target on an assumed identical cohort. Its Table IV test support is 7,041, whereas 15% of the original HAM10000 images is about 1,502; the paper's text differs on whether oversampling occurred before or after partitioning. Without its code and image IDs, the exact sequence cannot be established. A possible duplicate crossing risk is an inference, not a proven finding about the authors' implementation.

The research gap addressed here is the interaction between attention, class imbalance, and evaluation quality under limited compute. Published percentages often differ in number of classes, balancing of the evaluation population, lesion overlap, or reuse of validation data. Our report therefore presents the split, cohort size, paired metrics, and selection status next to every result.

The original dataset article [1] provides the common task and the reason lesion identities matter. Pardede et al. [2] provides the closest seven-class EfficientNet-B0 comparison. Haque et al. [3] suggests that greater resolution and a much larger EfficientNetV2 attention model can be promising, while its

unverified lesion grouping and larger compute budget limit a direct comparison. PanDerm [4] offers dermatology-pretrained features as a research probe, but its transformer architecture does not satisfy the approved final EfficientNet family. The selected three-category ensemble paper [5] and public EfficientNet-B2 implementation [6] further illustrate how class count, split design, augmentation, and training scale change an apparent benchmark.

What the Literature Contributed to Our Design

The dataset paper [1] established the seven diagnostic labels and the fact that images can share lesion identities. That observation shaped our decision to use lesion-grouped development partitions. The base paper [2] gave a directly relevant EfficientNet-B0 and class-balancing starting point, but it does not describe CBAM as part of its compared models. Our attention module is therefore an extension in architecture, while the separate Grad CAM procedure plays the same explanatory role used in related work. The large EfficientNetV2-L study [3] suggested that scale and resolution might matter; our inference-only resolution checks were deliberately much cheaper than reproducing its full training schedule.

The literature review also changed how we interpret a published percentage. For the same dataset name, investigators can use three or seven categories, different train-validation-test boundaries, repeated images, or balanced evaluation populations. The three-category ensemble [5] addresses a reduced task. The public EfficientNet-B2 project [6] uses an image-level split, which is a useful software example but has a different independence guarantee from our lesion-disjoint split. We therefore compare papers for design ideas and protocol details first, then discuss their reported performance with the correct qualification. This makes the review useful to our experimental decisions rather than a list of large numbers.

4 Research Gap

The literature establishes that transfer learning, balancing, attention, and ensembles can perform well on dermoscopic data, but a headline result often omits the distinctions needed to explain why it differs from another study. The selected base paper’s Table IV contains 7,041 balanced test cases despite a stated 70/15/15 split of 10,015 original images. Its public methods text does not settle the order of oversampling and partitioning. Our audit supports a comparability concern, not a definitive accusation about its implementation. The project therefore treats evaluation traceability as part of the research method.

Within this setting, the project studies a compact EfficientNet-B0 plus CBAM classifier and systematically checks the effect of imbalance interventions and inference choices. It also asks whether gains obtained under an image-level development split or with an alternative pretrained backbone carry over to a lesion-disjoint protocol. Maintaining these boundaries is a concrete methodological contribution because it prevents a stronger but differently obtained validation figure from being presented as the final model result.

5 Proposed System and Methodology

The implemented core model uses ImageNet pretrained EfficientNet B0 features, a single CBAM block after the final spatial feature map, global average pooling, and a seven output classifier. CBAM first learns which feature channels to emphasize and then which spatial positions to emphasize. This attention acts during the forward pass. Grad CAM, implemented as a separate explanation procedure, uses gradients from a selected class to produce a heatmap after prediction; it is not a balancing method or a direct accuracy booster. The present results do not establish that the model consistently attends to medically meaningful areas, so qualitative review is still required.

Original 600 by 450 images are resized to 224 by 224, converted to tensors, and normalized with ImageNet statistics. The baseline training augmentation is a horizontal flip. AdamW uses a 3e-5 backbone learning rate and 1e-4 for CBAM and head, batch size 64, mixed precision, and a 20 epoch cap with early stopping. The best checkpoint maximizes validation macro F1. The local best and latest checkpoints,

epoch histories, configurations, metrics, and an experiment ledger are retained. The first weighted run's final ledger update failed after training; its valid artifacts and CSV row were repaired without retraining. The baseline imbalance intervention weights cross entropy using square root inverse training class frequency. A separate experiment oversampled training references to the majority class count; it did not copy images into validation or test. A focal loss run tested an alternative emphasis on difficult examples. This creates controlled comparisons inside the same core architecture. Later inference and feature probes reused saved checkpoints where possible.

Data Preparation and Preprocessing

Metadata was linked to original images by image ID and used to verify the seven labels and lesion IDs. Images are RGB and are resized to 224 by 224 for the core run, transformed to tensors, then normalized using ImageNet channel statistics. The baseline training pipeline adds a horizontal flip with probability 0.5; validation preprocessing is deterministic. Class frequencies for weighting and oversampling come only from training rows. Keeping validation and test at their original frequencies lets their metrics describe a natural, unbalanced cohort rather than a synthetic balanced population.

EfficientNet, Transfer Learning, and Classification Head

The core backbone is EfficientNet-B0 with ImageNet-pretrained weights. Transfer learning begins from features learned on a large general-image collection and adapts them to dermoscopy, which is more feasible than learning all visual features from scratch with 10,015 task images. Its final feature map has 1,280 channels. After CBAM, adaptive global average pooling produces a compact vector, followed by a project-defined classifier with 512 and 128 hidden units and seven output scores. These head dimensions are implementation choices and are not attributed to the base paper.

The seven scores are converted into class probabilities for evaluation and probability averaging. The model is optimized end-to-end with AdamW. The backbone learning rate is lower than that of the new CBAM and head layers, reflecting that pretrained features need smaller updates while new layers must learn the task. The run uses mixed precision and batch size 64. The maximum is 20 epochs, with a plateau-based learning-rate reduction and early stopping after non-improving validation macro F1. The best checkpoint is therefore the best observed macro-F1 epoch, not necessarily the last epoch or the highest-accuracy epoch.

CBAM and Grad CAM

CBAM receives the final spatial features. Its channel branch combines average and maximum summary maps with a shared small network to learn which feature channels to strengthen or suppress. Its spatial branch combines channelwise average and maximum maps and applies a 7 by 7 convolution to learn where to emphasize. Because the block follows the final backbone stage, its spatial grid is relatively coarse. Placing attention earlier or in several stages might be investigated later, but no such trained result is claimed here.

Grad CAM is a separate explanation procedure. For a chosen class, it uses gradients at a convolutional feature map to create a heatmap of image regions associated with that class score. The repository implements heatmap generation, but the project has not yet finished a systematic explanation review. A bright area can be influenced by background or artifacts and is not proof that the model identified a medically valid sign. The final analysis should show representative correct and incorrect examples and describe these limits.

Imbalance Handling and Model Selection

The weighted baseline assigns larger cross-entropy weights to less frequent training classes using a square-root inverse-frequency rule, normalized to mean one. This moderates the extreme nevus-to-dermatofibroma count ratio. In a controlled alternative, random oversampling repeats training references to reach the majority count for every class, with fresh random training flips when an image is loaded. Focal loss was tested in another run to reduce the contribution of easy examples. These interventions were compared using validation macro F1 as well as accuracy, because each can shift the balance between majority and minority errors.

Why the Method Controls Evaluation Leakage

The fixed grouped split is stored as a manifest, so a training script reuses the same image assignments instead of silently producing a more favorable partition. Grouping is applied before training-only weights or repeated references are derived. Neither oversampling nor validation-based model selection can change which original images belong to validation or the locked test set. This design addresses the concrete repeated-lesion risk in HAM10000; it does not guarantee generalization to other clinics or devices. The exploratory image-level partition was created from the development pool for a separate protocol question and is explicitly not treated as equivalent evidence.

Why Accuracy and Macro F1 Are Reported Together

Accuracy counts every correct image equally and is sensitive to the large nevus category. Macro F1 first evaluates the precision-recall balance within each category and then averages the seven category scores equally. Consequently, a decision rule that shifts predictions toward nevus can increase accuracy while reducing macro F1 or minority recall. The saved confusion matrices and per-class reports are needed to interpret this trade-off, particularly for melanoma and visually similar benign categories. Checkpoint selection uses macro F1, while both metrics are reported for every leading configuration.

6 Implementation Overview

The repository contains modules for data preparation, model construction, training, metrics, evaluation, and Grad CAM, plus saved configurations and scripts for subsequent validation probes. Each completed CNN run has a best and latest checkpoint, history, and validation metrics. The experiment CSV records trained variants; further comparisons have result summaries and protocol notes. The original weighted run finished all 20 epochs but crashed during its final ledger update. Its checkpoint and metrics were valid, and the ledger row was repaired without retraining. This illustrates why independent artifact checks matter when bookkeeping fails after training.

Large raw images, local checkpoints, third-party PanDerm weights, and intermediate feature caches are excluded from the public Git repository, while code and numerical summaries remain. The exploratory branch keeps the original grouped benchmark distinct from later image-level experiments. This organization lets the final report identify which model produced each result without mistaking a cached research probe for the approved final model.

7 Experiments Performed

Core EfficientNet B0 plus CBAM comparisons on lesion separated validation

Method	Accuracy	Macro F1	Observation
Weighted cross entropy	84.56%	0.7569	Best original trained checkpoint for balanced metrics; best epoch 16.
Training only random oversampling	85.30%	0.7321	Accuracy rose; macro F1 fell and repeated image overfit appeared.
Focal loss	83.77%	0.7238	Did not improve the prior leaders.
Weighted plus focal probability average	84.36%	0.7718	Highest pre exploration strict macro F1; selected on validation.
Oversampled TTA plus prior correction	85.89%	0.7227	Highest pre exploration strict accuracy; more majority class predictions weakened balance.

Horizontal flip test time augmentation improved the weighted run's macro F1 to 0.7704 while lowering its accuracy to 84.23%. Probability ensembles yielded small gains in macro F1. A frozen post CBAM embedding classifier, metadata only or metadata fusion, MC dropout, and a center crop diagnostic did not improve both main metrics. The metadata probe did not clear its preset improvement gate, so full multimodal CNN training was not pursued. Prior correction modestly raised accuracy while reducing

melanoma performance and macro F1. These are useful negative and tradeoff findings, not discarded work.

Validation-Only Inference and Feature Studies

The first no-training question was whether averaging predictions across the original image and a horizontal flip could reduce isolated errors. Weighted B0 macro F1 rose from 0.7569 to 0.7704, while accuracy fell from 84.56% to 84.23%. An oversampled flip average reached 85.76% accuracy but only 0.7296 macro F1. A small weighted-plus-focal probability blend reached the pre-exploration strict macro-F1 leader of 0.7718. These experiments showed complementarity but also that the best configuration depends on which metric is selected.

The next question was whether post-CBAM embeddings or non-image metadata could add information cheaply. A frozen embedding logistic classifier reached 83.83% strict accuracy and 0.7347 macro F1, below the CNN leaders. Age, sex, and lesion location alone were weak; the best metadata fusion did not pass the prespecified improvement gate, so full multimodal CNN training was not justified. MC dropout averaged repeated stochastic passes but also missed the leaders. A center-crop diagnostic reduced accuracy to 81.64%, and prior correction increased oversampled accuracy to 85.89% while lowering macro F1 to 0.7227. These results steered work away from expensive retraining based on weak inference gains.

Exploratory split, resolution and augmentation

On the image level development split, a weighted B0 plus CBAM run reached 86.23% accuracy and 0.7808 macro F1. A stronger augmentation run using flips, rotation, affine movement, and mild color jitter reached 85.30% and 0.7657 on the same validation cohort. It reduced training memorization but did not improve validation performance. Rather than start another long training run, we averaged probabilities from the two saved B0 checkpoints at 224 and 384 pixels. The four stream average reached 88.16% and 0.8280 macro F1. A comparable multi resolution check on strict validation did not beat the earlier strict leaders, so the gain is not presented as a general result.

The image-level protocol comparison was intended to test whether split choice could explain part of the gap from published scores. A weighted B0 plus CBAM run on that split reached 86.23% accuracy and 0.7808 macro F1, modestly above the original weighted checkpoint's numbers on a different strict cohort. It did not approach 93%. We then changed only training augmentation, adding vertical flips, rotation, affine movement, and mild color jitter. That stronger recipe reduced training memorization but its best checkpoint fell to 85.30% and 0.7657 on the same exploratory validation cohort, so it was not selected.

Before training again, saved predictions were examined at a larger 384-pixel input. A checkpoint evaluated only at 384 pixels was weaker than at 224, yet averaging 224 and 384 predictions improved its result, suggesting complementary errors. Four equal streams from two saved B0 checkpoints at both resolutions reached 88.16% accuracy and 0.8280 macro F1 on exploratory validation. The same principle did not improve strict validation leaders, so the conclusion is limited to the image-level cohort. It is an inference improvement using existing weights, not evidence that a newly trained larger input model would perform better.

Domain pretrained feature probe outside the proposed final architecture

The PanDerm research sequence began with frozen dermatology-pretrained features and shallow heads on the grouped development split. Its RBF SVM alone achieved 84.63% strict accuracy. A two-epoch partial fine-tuning pilot reached 84.43% and was stopped under a preset threshold instead of consuming a full run. B0 and PanDerm errors were partly complementary, producing an 87.62% strict accuracy four-stream candidate. An orientation-view check gave 87.69% strict accuracy and 0.7771 macro F1. The strongest strict macro F1, 0.7837, came from another PanDerm/B0 combination at 87.03% accuracy.

The image-level PanDerm probe used four deterministic orientation views and the existing four B0 resolution streams. Their eight-way equal probability average reached 90.75% accuracy and 0.8573 macro F1 on exploratory validation, a 2.59-point improvement over its four-B0 control on the same 1,503 images. Because 596 validation images share a lesion with training and the candidate was selected among

validation variants, it is not an independent estimate for new lesions. Because it contains PanDerm, it is not the approved final EfficientNet plus CBAM model. Both qualifications accompany the result wherever it is summarized.

Experiment Rationale and Stop Decisions

The experiment sequence moved from controlled training comparisons to lower-cost diagnostic checks. The oversampling run asked whether repeatedly exposing minority training images would outperform the original weighted-loss checkpoint; the fall in macro F1 and large training-validation gap argued against selecting it for balanced performance. Focal loss asked whether reducing the contribution of easy cases would help; it did not surpass either core leader. Rather than repeatedly retrain the same backbone, later checks reused saved probabilities or post-CBAM features. Metadata fusion, MC dropout, cropping, and prior correction failed to produce a convincing paired gain, so the project did not escalate each idea into a full CNN training run.

The image-level probe was designed to test a protocol hypothesis using the same B0 plus CBAM architecture, not to claim a clean improvement over the grouped benchmark. Stronger training augmentation was then evaluated on that same exploratory cohort and produced lower accuracy and macro F1 than its baseline. The multi-resolution average made use of those preserved checkpoints without another training run. PanDerm was finally tested because dermatology-pretrained features might make different errors from ImageNet-trained B0. Its short fine-tune pilot was stopped after two epochs under a preset threshold, while frozen features and orientation views were retained for clearly marked exploratory ensembles.

8 Results and Discussion

Protocol and configuration	Accuracy	Macro F1	Interpretation
Strict lesion separated: weighted B0 plus CBAM checkpoint	84.56%	0.7569	Core trained baseline.
Strict lesion separated: best accuracy PanDerm/B0 ensemble	87.69%	0.7771	Best strict numeric accuracy; outside final architecture.
Strict lesion separated: best macro F1 PanDerm/B0 combination	87.03%	0.7837	Best strict macro F1, different configuration.
Exploratory image level: four B0 plus CBAM streams	88.16%	0.8280	Best compatible exploratory ensemble; shared lesions.
Exploratory image level: four B0 plus four PanDerm streams	90.75%	0.8573	Best overall numeric validation accuracy; shared lesions and outside final architecture.

The highest accuracy and highest macro F1 on the strict split come from different configurations. Class imbalance makes this distinction essential. The exploratory 90.75% result is a measured improvement over the same split's four stream B0 control by 2.59 percentage points, but it is validation selected and does not establish 90.75% performance on unseen lesions. No full cohort seven class score above 93% has been obtained. No locked test evaluation has been run.

Several findings guide the remaining work. Training only oversampling improved accuracy but weakened class balance. Stronger augmentation was not automatically better. Resolution diversity helped one exploratory cohort but not the lesion separated cohort. Dermatology pretrained features complemented B0 predictions, yet the resulting method conflicts with the agreed final architecture. We will therefore distinguish exploratory evidence from the final EfficientNet plus CBAM study and avoid choosing the final method using the locked test set.

Reading the Main Result Table

The 87.69% lesion-disjoint validation accuracy and 0.7837 lesion-disjoint macro F1 are separate maxima: the latter belongs to a configuration with 87.03% accuracy. The 88.16% four-stream B0 result is the best EfficientNet plus CBAM exploratory image-level ensemble, whereas 90.75% is the largest numeric full-validation accuracy from the PanDerm-assisted eight-stream candidate. Because the image-level

development split shares lesions and those candidates were selected after validation comparisons, neither percentage can be relabeled as a held-out test estimate. The locked test cohort has not been scored. This distinction is central to the result, not a footnote.

The experiment records also reveal that a small improvement in one metric may correspond to a change in error type rather than better discrimination across every class. The prior-corrected oversampled setting, for example, gains accuracy while weakening macro F1. The strict multi-resolution check did not transfer the exploratory ensemble gain. These observations limit the conclusion we can draw from any single technique: a method is selected only after considering its metric pair, classwise behavior, compute cost, architecture fit, and evaluation protocol.

9 Comparison of Techniques Tried

The three original B0 plus CBAM training runs isolate different responses to class imbalance. Weighted cross entropy retained the strongest balanced checkpoint among the individually trained core models. Oversampling raised overall accuracy but reduced macro F1 and showed a larger training-to-validation gap. Focal loss did not surpass either leader. Therefore a different loss function alone has not solved the minority-class problem. The macro-F1 checkpoint selection rule remains useful because it exposes this trade-off even when a high accuracy setting appears attractive.

Inference changes were less expensive to test than another 20-epoch run. Flip TTA and a small probability blend produced modest macro-F1 gains. Frozen-feature classification, metadata fusion, MC dropout, cropping, and prior correction either missed existing leaders or improved accuracy at the expense of balanced performance. Multi-resolution averaging improved the exploratory cohort but not the grouped one. The PanDerm study obtained the largest numeric gains, showing complementary domain-pretrained features, yet changed the architecture. The lessons are conditional: a method can help one metric, one cohort, or one architecture without becoming the final selection.

10 Important Findings and Lessons Learned

First, score reporting must include both accuracy and macro F1 because a majority-class shift can raise one and lower the other. Second, validation-selected ensembles are development findings even when the validation images are lesion-disjoint. Third, an image-level split with repeated-lesion overlap has a weaker independence guarantee. Fourth, stronger augmentation and higher resolution cannot be assumed to help simply because they are common in literature; the actual matched comparisons matter. Fifth, preserving checkpoints made cheap inference checks possible and saved further GPU training. Finally, negative results clarified which hypotheses did not justify more compute.

The error analysis still points to confusions among visually similar categories, especially melanoma, benign keratosis, and nevus. The available experiments do not establish that an attention block solves those confusions, since a controlled B0 without CBAM ablation has not been reported. The current implementation demonstrates a functioning attention-equipped classifier, while the isolated contribution of CBAM remains an open empirical question for the final paper. This distinction keeps the research claim proportional to the evidence.

11 Comparison with Published Work

Pardede et al. [2] report a much larger percentage, but their 7,041-case balanced test table and ambiguous balancing order make the evaluation population different from an untouched original-image 15% test. The three-category 96% study [5] solves a smaller classification task. The 91.15% EfficientNetV2-L attention work [3] uses a much larger model, higher input resolution, and a split whose lesion independence is not established in the inspected material. The public B2 implementation [6] uses image-level splitting and reports 82.24% test accuracy. These comparisons informed hypotheses about scale, augmentation, and protocol, but none supplies a like-for-like target for our strict seven-class split. The project's 90.75% result is an exploratory validation number, not a claim of superiority over a paper's final test result. Conversely, the 87.69% grouped validation accuracy cannot be treated as definitive generalization without the reserved test assessment. The final report should tabulate class count, lesion

grouping, balancing order, validation use, and test cohort beside published accuracy wherever those facts can be verified. Unreported details should be marked unknown rather than filled by assumption.

The reported 98.86% from the base paper [2] cannot be used as a direct pass/fail target for our untouched original-image or grouped validation because its published test table totals 7,041 balanced cases. The public description leaves the exact balancing and split sequence unresolved. We report this as a methodological uncertainty and possible source of inflated comparability, not proof of a particular implementation error. Likewise, the larger EfficientNetV2-L result [3] and selected three-category result [5] answer related but differently defined questions. A sound final comparison will show task size, partition identity, validation selection, and any known overlap beside each metric.

12 Current State of the Project

The core dataset preparation, fixed grouped split, EfficientNet B0 plus CBAM training pipeline, checkpointing, validation analysis, and Grad CAM generation code are complete. The repository contains saved training records and reproducible exploratory comparisons. The original test partition remains locked. A final model configuration has not yet been frozen, a systematic review of explanation heatmaps is pending, and no final test result or clinical validation claim is available.

- Select a final configuration consistent with the approved EfficientNet, CBAM, and Grad CAM methodology using development evidence only.
- Review representative Grad CAM heatmaps across correct and incorrect cases, including minority classes; state qualitative limits.
- After model and reporting rules are fixed, evaluate the original untouched test partition once and report per class metrics and confusion matrix.
- Write the final paper with protocol specific tables, literature comparison, limitations, and reproducibility details.

At this stage, the core classifier, grouped data manifest, training pipeline, saved checkpoints, validation summaries, and Grad CAM generation path exist. The project has not yet demonstrated an isolated performance gain caused by CBAM, and it has not completed a systematic explanation audit. Its largest numeric validation candidate includes PanDerm and is not eligible as the approved final architecture. The remaining decision is therefore to lock a configuration within the EfficientNet, CBAM, and Grad CAM scope using development evidence, then assess it once on the reserved test partition. The final paper can report exploratory methods as research findings without confusing them with the final model.

13 Remaining Work and Future Work

Near-term work is methodological rather than an open-ended search for larger numbers. The team must select a final method inside the approved EfficientNet, CBAM, and Grad CAM scope using the development evidence, document its full configuration, and freeze the decision before accessing the original locked test images. A systematic Grad CAM review should include correct predictions, errors, and minority-class examples. The heatmaps should be discussed as visual evidence with possible artifacts, not treated as proof of clinical correctness.

Once the model and reporting plan are fixed, a single final evaluation can produce the held-out accuracy, macro F1, per-class precision and recall, and confusion matrix. The final paper should then contrast this result with validation findings and relevant published protocols, discuss class imbalance and selection bias, and state the limitation of one dataset and no external clinical validation. Additional EfficientNet variants or CBAM placement ablations may be future work, but no unrun experiment is described here as completed.

14 Conclusion

The project has progressed beyond a proposed design to a working, documented seven class classifier and a substantial validation study. EfficientNet B0 plus CBAM is the implemented core, while the largest measured validation percentage comes from a separate PanDerm assisted image level exploration. The

main scientific conclusion to date is that architecture, balancing, and evaluation protocol must be reported together. Final model selection and held out assessment remain the decisive next steps.

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